

Energy-Efficient 200 Gbit/s Transceivers for Optical Access Networks

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Chapter 1

Access Networks and Coherent PON

1.1 Optical Access Networks

The access network is the final link between an operator's central exchange and the end user, often called the *last mile*. Exponentially growing bandwidth demands from streaming, cloud services, and 5G backhaul are placing severe pressure on this segment.

A passive optical network (PON) uses a point-to-multipoint (P2MP) topology in which a single optical line terminal (OLT) at the exchange serves multiple optical network units (ONUs) at the customer premises through a tree of passive optical splitters, as illustrated in fig. 1.1. No active components are required in the distribution plant, keeping operational costs low.

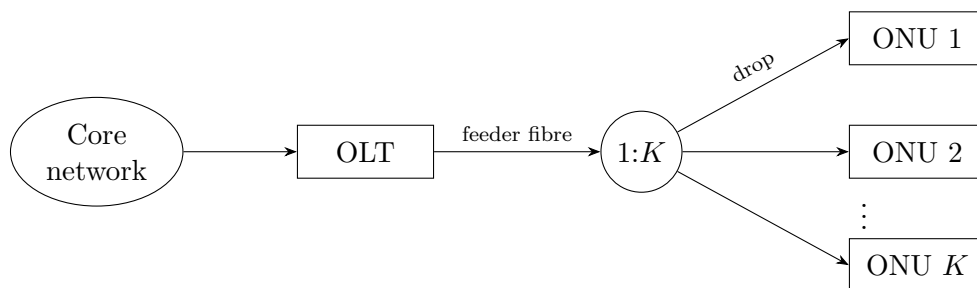


Figure 1.1: Passive optical network: one OLT serves K ONUs via a passive $1:K$ splitter.

1.2 Coherent PON

Traditional intensity-modulation direct-detection (IM-DD) passive optical networks are reaching their practical limits at data rates around 50 Gbit/s per wavelength. Coherent technology encodes information in the amplitude and phase of the optical signal, allowing modulation formats like QPSK that can carry more bits per symbol. Coherent receivers are also more sensitive, allowing for lower transmit powers. These factors make them a promising alternative to achieve rates of 200 Gbit/s on an acceptable power budget [7].

1.3 Specification

The Coherent Passive Optical Network (CPON) specification [1] defines a single-wavelength P2MP system and serves as the reference specification for the receiver implementations evaluated in this report. This report focuses on the *downstream* direction (OLT to ONU) where receiver optimisations can be shared by all ONUs; the key physical-layer parameters are described in the following sections.

1.3.1 Modulation and Symbol Rate

The downstream signal uses non-differential dual-polarisation QPSK (DP-QPSK) at a symbol rate of $R_s = 30.5$ GBd. Each polarisation carries 2 bits per symbol by mapping consecutive bit pairs to in-phase and quadrature amplitudes $I, Q \in \{-1, +1\}$, giving QPSK constellation points at $\pm 1 \pm j$. Training and pilot symbols use the same QPSK constellation as the data symbols. Soft-decision forward error correction (FEC) is used with a 22% coding overhead.

1.3.2 Downstream Frame Structure

The fundamental framing unit is the *DSP subframe* of 3 712 symbols, organised as 116×32 -symbol blocks. Every block begins with a pilot symbol; the first block also carries an 11-symbol training sequence, whose first symbol doubles as pilot index 1.

1.3.3 Key System Tolerances

The key physical-layer tolerances that the receiver DSP must accommodate are summarised in table 1.1. The transmitter SNR of 35 dB is the back-to-back signal quality at the OLT, accounting for impairments in the core network before the access link, and sets the noise floor NSR_0 used in the transmitter-energy model of chapter 2. The worst case values of table 1.1 are used in each experiment.

Table 1.1: Key downstream tolerances from the CPON specification [1].

Parameter	Value
Loss budget (OLT to ONU)	35 dB
Transmitter SNR	35 dB
Worst-case CFO	≈ 3 GHz
Laser linewidth (OLT and ONU)	≤ 1 MHz
Sampling frequency offset (SFO)	≤ 40 ppm
Chromatic dispersion	≤ 1600 ps/nm
Polarisation-mode dispersion	≤ 8 ps
Reach	≤ 80 km

The specification requires a pre-FEC bit error ratio of $\leq 2 \times 10^{-2}$, which the soft-decision LDPC FEC reduces to a post-FEC BER of $\leq 10^{-12}$.

Chapter 2

Network and Receiver Models

2.1 Simulation Framework

All DSP algorithms in this report are designed, characterised, and compared through software simulation in MATLAB. The simulated receiver processes DP-QPSK symbols generated at the CPON downstream parameters described in chapter 1. The receiver follows the standard coherent-DSP pipeline shown in fig. 2.1.

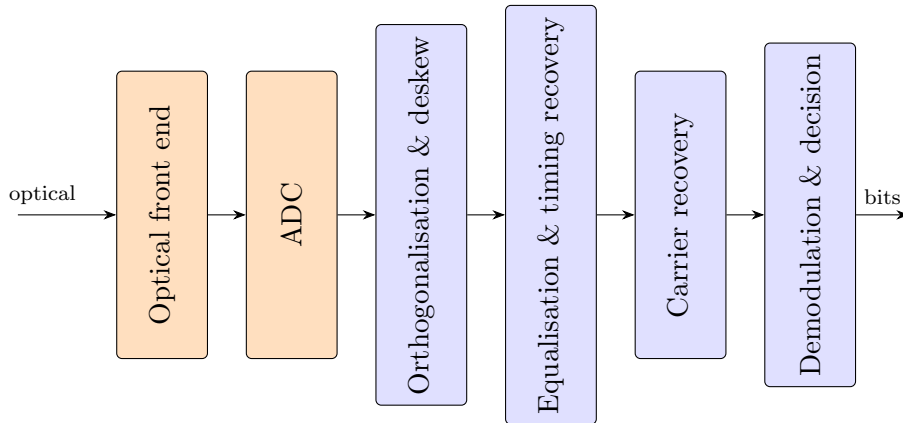


Figure 2.1: DSP pipeline of the simulated coherent receiver. The analogue stages (orange) and the digital stages (blue) are distinguished by colour. The optical front end mixes the received signal with a local-oscillator laser to produce four baseband photocurrents, which the ADCs digitise; orthogonalisation and deskew correct front-end imbalances; equalisation and timing recovery compensate the linear channel and recover the symbol-rate sampling phase; carrier recovery removes the residual CFO and laser phase noise; the final stage maps each symbol to bits.

2.1.1 Fixed-Point Arithmetic

The receiver is evaluated with fixed-point arithmetic, as used in ASIC designs. This captures the effect of quantisation error on each implementation's performance, and underpins a bit-accurate energy model that goes beyond algorithmic complexity. All fixed-point modelling uses the MATLAB Fixed-Point Designer toolkit.

Fixed-point values require a well-defined range that they represent. The range of the input signal can vary due to additive noise and dispersive effects so normalisation to the range $[-1,1]$ on the real and imaginary axes is first performed before further processing. The normalisation must balance some points exceeding the range and saturating with points being compressed to the centre. A suitable approach found by trial-and-error is the linear rescaling by the signal magnitude at the 99.9th percentile with larger amplitudes saturating.

2.2 Energy Estimation

The energy-efficiency of a receiver implementation cannot be determined from the receiver energy alone. A simpler architecture may reduce receiver energy but, through a drop in accuracy, incur an SNR penalty that raises the transmit power needed for acceptable operation. This calls for a system-level model of the energy consumption where an SNR penalty can be converted to an equivalent increase in transmitter energy, allowing receiver implementations to be compared by their combined energy.

2.2.1 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. The reference values shown in table 2.1 provide a useful estimate to compare DSP algorithms but a more accurate estimate would require power simulations for an ASIC implementation.

Table 2.1: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [9].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

Dependence on Word Length A ripple-carry adder performs n bit-wise additions on words of length n , so its energy scales linearly with n . A shift-and-add multiplier accumulates n partial products of length n and so scales as n^2 . Modern adders and multipliers (e.g. carry-lookahead, Booth/Wallace) reduce these costs by a constant factor or to $n \log n$, so the scalings used here are upper bounds; comparisons between algorithms remain valid because all are penalised equally. The proportionality constants are calibrated by a least-squares fit through the origin to the two data points of table 2.1, giving

$$E_A(n) = 3.16 n \text{ fJ}, \quad E_M(n) = 3.03 n^2 \text{ fJ}. \quad (2.1)$$

Energy consumption for arithmetic operations in modern process nodes is not widely reported, and Dennard scaling (the rule that transistor power density remained constant as they were scaled down) [5] can no longer be applied to extrapolate from the values in table 2.1. As an approximation, the $\approx 75\%$ reduction in total power dissipation at constant frequency reported for an Intel core moving from 45 nm to 14 nm CMOS [18] is

taken as a proxy for the per-operation energy scaling. This conflates contributions from leakage, clocks, caches and I/O with the arithmetic energy of interest, so the resulting numbers should be read as an estimate rather than a calibrated figure. Applying this reduction to the 45 nm fit gives

$$E_A(n) = 0.79 n \text{ fJ}, \quad E_M(n) = 0.76 n^2 \text{ fJ}. \quad (2.2)$$

Amortisation Some DSP algorithms perform computations that are reused over multiple symbols. For example, carrier frequency offset (CFO) varies much slower than the symbol rate so can be estimated less frequently. For a computation that requires N_M multiplications and N_A additions and is used by S symbols, the operations per symbol would be N_M/S and N_A/S .

CORDIC Coherent receivers perform phase estimates and rotations on complex-valued data in several DSP blocks. This is done efficiently in fixed-point hardware with the CORDIC algorithm, which evaluates these operations using only shifts and additions. CORDIC iteratively rotates a vector by the fixed angles $\arctan(2^{-i})$, each implemented as a bit-shift and an add/subtract. Each iteration adds roughly one bit of angular resolution, so n bits of precision require n CORDIC iterations and the cost is taken to be equivalent to an n -bit real multiplication.

Total Energy and Power Many DSP blocks operate on complex-valued data; each complex multiplication is counted as 4 real multiplications and 2 real additions, and each complex addition as 2 real additions. The energy per symbol for the entire receiver is then

$$E_{\text{rx},s} = \sum_{k=1}^L 2 [N_{A,k} E_A(n_k) + N_{M,k} E_M(n_k)], \quad (2.3)$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarisations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s . The receiver power is then

$$P_{\text{rx}} = R_s E_{\text{rx},s}, \quad (2.4)$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\text{rx}} = \frac{E_{\text{rx},s}}{\log_2(M)}, \quad (2.5)$$

where M is the modulation order.

2.2.2 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 1.1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share.

Minimum Required SNR The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 2×10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR (RSNR) to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

Noise The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power referred to the optical input of a balanced coherent receiver is $2h\nu B_{\text{eff}}$, where the factor of 2 captures the contributions from the signal and LO arms of the balanced detector summed in quadrature, and B_{eff} is the receiver electrical bandwidth, set equal to the symbol rate by a matched bandpass filter after the 1:K splitter in fig. 1.1. The power per polarisation received by each ONU is $P_{\text{tx}}/2\Gamma K$, where Γ is the loss from OLT to ONU. This gives a per-ONU NSR of

$$\text{NSR} = \frac{4\Gamma K h\nu B_{\text{eff}}}{P_{\text{tx}}} + \text{NSR}_0, \quad (2.6)$$

where $\text{NSR} = 1/\text{SNR}$ and NSR_0 is the back-to-back NSR at the OLT. Rearranging gives the transmitted power as

$$P_{\text{tx}} = \frac{4\Gamma K h\nu B_{\text{eff}}}{\text{NSR} - \text{NSR}_0}. \quad (2.7)$$

Assuming Nyquist pulse-shaping and matched bandpass filtering at the receiver ($B_{\text{eff}} = R_s$), the energy per bit transmitted is

$$E_{\text{tx}} = \frac{P_{\text{tx}}}{2K R_s \log_2(M)} = \frac{2\Gamma h\nu}{(\text{NSR} - \text{NSR}_0) \log_2(M)}. \quad (2.8)$$

2.2.3 Combined Energy

Before calculating a combined energy contribution, the transmitted optical energy in eq. (2.8) must be converted into an energy drawn from the electrical power supply by estimating the efficiencies of the transmitting laser and the modulator. The dual-polarisation modulator (fig. 2.2) consists of two nested-Mach-Zehnder IQ modulators sharing a common laser through a polarisation beam splitter, with their outputs recombined by a polarisation beam combiner. Each IQ modulator has the transfer function [16]

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \frac{1}{2} \sin \left[\frac{v_I(t)\pi}{2V_\pi} \right] + \frac{j}{2} \sin \left[\frac{v_Q(t)\pi}{2V_\pi} \right], \quad (2.9)$$

where each MZM is biased at its transmission null so that $v_I = v_Q = 0$ produces no output and the drive voltages swing symmetrically about 0 V.

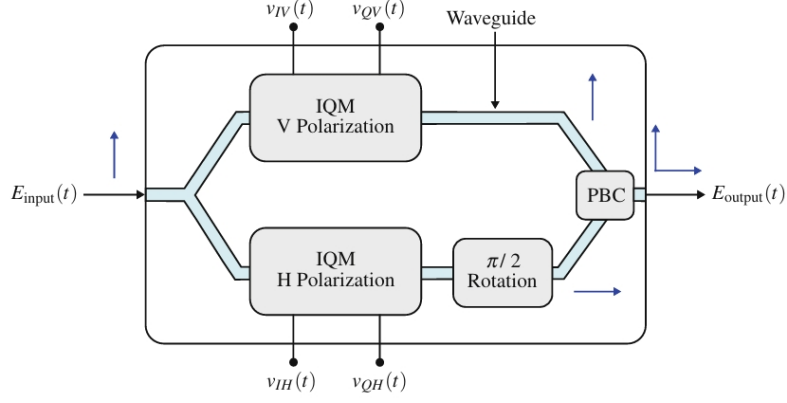


Figure 2.2: Dual-polarisation coherent modulator using MZMs [16].

QPSK is generated by driving each MZM between adjacent transmission peaks, $v_I(t), v_Q(t) \in \{-V_\pi, +V_\pi\}$, so

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \pm \frac{1}{2} \pm \frac{j}{2} \quad (2.10)$$

and $P_{\text{out}} = P_{\text{in}}/2$, where the loss comes from power coupled to unused ports in the 2×2 couplers. An ideal polarisation beam splitter and combiner introduce no further loss, so the *marginal* efficiency of the dual-polarisation modulator is

$$\eta_{\text{DPM}} = \frac{dP_{\text{tx}}}{dP_{\text{in}}} = 0.5. \quad (2.11)$$

The overall efficiency would also include the power dissipated in the drive electronics, but this is independent of transmit power and so cancels in a comparison of receiver implementations. Combining with the wall-plug efficiency of the laser η_{WP} gives an overall efficiency of

$$\eta_{\text{T}} = \eta_{\text{WP}} \eta_{\text{DPM}} = 0.5 \eta_{\text{WP}}. \quad (2.12)$$

Recently reported values for the wall-plug efficiency of lasers suited to coherent systems are 16-19% [21, 19] which gives an overall efficiency of just under 10%. A measure of combined energy is then

$$E_{\text{sys}} = \frac{1}{\eta_{\text{T}}} E_{\text{tx}} + E_{\text{rx}}. \quad (2.13)$$

This does not give the total energy consumed by the optical network, but a difference in E_{sys} shows the energy savings across receiver implementations.

Chapter 3

Equalisation and Clock Recovery

3.1 Introduction

The receiver must compensate two distinct channel impairments and recover the sampling phase before symbol decisions are made. Chromatic dispersion (CD) is static and accurately characterised by the fibre parameters, so it is removed by a fixed filter derived analytically from the channel model. Polarisation-mode dispersion (PMD) varies on millisecond timescales and is unknown a priori, so it is tracked by an adaptive equaliser that also absorbs residual inter-symbol interference (ISI) from the transmit pulse shape. Independently, the receiver samples the incoming waveform with an unsynchronised clock that drifts relative to the transmitter, leaving a fractional timing error that must be estimated and corrected.

The static, adaptive, and clock-recovery blocks share the same data stream and can be arranged in several ways. A Gardner timing error detector (TED) operating in the time domain naturally pairs with a time-domain CD filter or directly with the adaptive equaliser; a Modified Godard TED (Godard hereafter) operates in the frequency domain and can re-use the same FFT as the overlap-save CD equaliser, eliminating the FFT cost of the timing recovery loop entirely. This chapter derives both equaliser stages and both timing recovery algorithms, then combines them into two end-to-end implementations that are compared on BER and energy. Both share an overlap-save CD/matched-filter front end and a single-tap adaptive equaliser, and differ only in the timing recovery: a Gardner digital phase-locked loop (DPLL) operating after the IFFT, or a Modified Godard loop embedded in the frequency domain.

Filter lengths and tap counts are sized for the worst-case CPON tolerances of section 1.3.3: 1600 ps/nm of CD, 8 ps of PMD, and a 30.5 GBd symbol rate at a central wavelength of 1550 nm.

3.2 Equalisation

3.2.1 Channel Impairments

Chromatic dispersion CD arises because the group velocity of light in fibre depends on wavelength, so the frequency components of a pulse accumulate a frequency-dependent

delay that spreads them over many symbol periods. It is an all-pass effect with a quadratic phase response, simulated by transforming the signal to the frequency domain, applying the transfer function

$$H_{\text{CD}}(\omega) = \exp\left(j \frac{D\lambda^2}{4\pi c} L \omega^2\right), \quad (3.1)$$

where D is the dispersion coefficient, λ the central wavelength, L the fibre length, and c the speed of light, then transforming back with an inverse FFT.

Polarisation-mode dispersion PMD is a frequency-dependent group delay between the two principal states of the fibre caused by random birefringence along the link, simulated as a concatenation of N_{pmd} birefringent sections, each applying a differential group delay (DGD) of $\pm\tau/2$ to its two principal states in the frequency domain, with a random unitary rotation (the singular vectors of a complex Gaussian matrix) modelling the mode coupling between sections. The mean DGD scales as $\tau \propto D_{\text{PMD}}\sqrt{L}$, with D_{PMD} the fibre PMD coefficient. PMD must be distinguished from *polarisation mixing*, the essentially frequency-flat unitary rotation of the polarisation pair that arises from any uncompensated rotation between the transmit and local-oscillator reference frames; the adaptive equaliser must track both, but only PMD requires multiple taps.

RRC pulse shaping The transmit pulse is a root-raised-cosine (RRC) with roll-off β and a finite truncation span; the matched filter at the receiver is identical to it, and the cascade of the two removes ISI and rejects noise. A larger β makes the time-domain pulse shorter and easier to truncate, allowing for fewer taps in the matched filter which reduces complexity. A span of 10 symbols with $\beta = 0.25$ is reasonable for an access network where bandwidth requirements are less strict and low complexity is critical.

3.2.2 Static Equalisation

Chromatic dispersion is compensated by applying the inverse of eq. (3.1),

$$H_{\text{eq}}(\omega) = \exp\left(-j \frac{D\lambda^2}{4\pi c} L \omega^2\right), \quad (3.2)$$

whose inverse Fourier transform is a finite-energy chirp

$$h(t) = \sqrt{\frac{c}{jD\lambda^2 L}} \exp\left(j \frac{\pi c t^2}{D\lambda^2 L}\right). \quad (3.3)$$

The equaliser may be realised either as a time-domain finite impulse response (FIR) filter per polarisation, with taps obtained by sampling eq. (3.3) at the receiver sampling period T_s and truncating to N_{CD} coefficients, or in the frequency domain. As the data stream is continuous, the frequency-domain filter is applied block-by-block by overlap-save: each block of N samples is transformed with an FFT, multiplied by the precomputed samples of H_{eq} , and inverse transformed. Discarding the first $N_{\text{CD}} - 1$ samples of every block removes the circular-convolution wrap-around, leaving $N - N_{\text{CD}} + 1$ valid output samples per block.

3.2.3 Adaptive Equalisation

PMD is tracked with a 2×2 multiple-input multiple-output (MIMO) equaliser comprising four FIR filters whose outputs are

$$x_{\text{out}} = \mathbf{h}_{xx}^H \mathbf{x}_{\text{in}} + \mathbf{h}_{xy}^H \mathbf{y}_{\text{in}}, \quad y_{\text{out}} = \mathbf{h}_{yx}^H \mathbf{x}_{\text{in}} + \mathbf{h}_{yy}^H \mathbf{y}_{\text{in}}. \quad (3.4)$$

The error terms are produced blindly using the constant modulus algorithm (CMA) [17], which exploits the fact that an ideal QPSK constellation lies on a circle of fixed radius. CMA minimises $J = \mathbb{E}[(|x_{\text{out}}|^2 - R)^2]$, where the dispersion constant $R = \mathbb{E}[|s|^4]/\mathbb{E}[|s|^2]$ reduces to a constant for QPSK. The stochastic gradient yields

$$e_x = R - |x_{\text{out}}|^2, \quad e_y = R - |y_{\text{out}}|^2, \quad (3.5)$$

and the filter weights are updated as

$$\begin{aligned} \mathbf{h}_{xx} &\leftarrow \mathbf{h}_{xx} + \mu e_x \mathbf{x}_{\text{in}} x_{\text{out}}^*, & \mathbf{h}_{xy} &\leftarrow \mathbf{h}_{xy} + \mu e_x \mathbf{y}_{\text{in}} x_{\text{out}}^*, \\ \mathbf{h}_{yx} &\leftarrow \mathbf{h}_{yx} + \mu e_y \mathbf{x}_{\text{in}} y_{\text{out}}^*, & \mathbf{h}_{yy} &\leftarrow \mathbf{h}_{yy} + \mu e_y \mathbf{y}_{\text{in}} y_{\text{out}}^*, \end{aligned} \quad (3.6)$$

with step size μ restricted to powers of two for hardware simplicity. CMA suffers from the singularity problem [17], in which both outputs converge to the same input; this is avoided by initialising $\mathbf{h}_{xx}$ with a single one at the centre tap and seeding the y -polarisation weights from the conjugate-symmetric relation $\mathbf{h}_{yx}[n] = \mathbf{h}_{xx}^*[-n]$, $\mathbf{h}_{yy}[n] = -\mathbf{h}_{xy}^*[-n]$ once x_{out} has converged [14]. CMA is also insensitive to phase rotations from CFO and phase noise, removing the need for a feedback path from the carrier-recovery block.

Parallelism. At a symbol rate of 30.5 GBd the equaliser must produce more than one output per CMOS clock period, so it is parallelised across P lanes. Each lane $p \in \{0, \dots, P-1\}$ owns its own input buffer

$$\mathbf{x}_p = [\mathbf{x}[mP + p], \mathbf{x}[mP + p - 1], \dots, \mathbf{x}[mP + p - (N_{\text{AEQ}} - 1)]] \quad (3.7)$$

and forms its own output and error using eqs. (3.4) and (3.5), but a single shared set of filter weights is held constant across the P lanes of a block. Each lane contributes its own gradient term and the weights are then updated once per block from the summed contribution; for $\mathbf{h}_{xx}$ this is

$$\mathbf{h}_{xx} \leftarrow \mathbf{h}_{xx} + \mu \sum_{p=0}^{P-1} e_x[p] \mathbf{x}_p x_{\text{out}}^*[p], \quad (3.8)$$

and analogously for the three other filters.

Sign-Sign CMA. A variant of the CMA update replaces the error and output terms by their signs: $e \rightarrow \text{sgn}(e)$ and $x_{\text{out}} \rightarrow \text{csgn}(x_{\text{out}})$, where the complex signum function is defined as

$$\text{csgn}(z) = \text{sgn}(\Re z) + j \text{sgn}(\Im z). \quad (3.9)$$

This removes all multiplications at no performance cost [2], so it will be employed in the later analysis.

Fixed-point The step size μ is typically $\approx 10^{-3}$. This means there must be different precisions for the filter weights and the gradient term $e_x \mathbf{x}_{\text{in}} x_{\text{out}}$. Choosing μ as a power of 2 allows the gradient to be quantised to a low precision then bit-shifted before updating the weights.

3.2.4 Matched Filtering

The receiver maximises signal-to-noise ratio at the symbol decision instants by filtering the received signal with a matched filter, the time-reversed conjugate of the transmit pulse. For the RRC pulse used here the matched filter is itself an RRC; the cascade of transmit and receive RRCs is a raised cosine that satisfies the Nyquist criterion. Two implementations are possible. The matched filter may be implemented explicitly as a frequency-domain mask multiplied into the overlap-save CD stage at no additional cost: $H_{\text{eq}}(\omega)$ is replaced by $H_{\text{eq}}(\omega)H_{\text{RRC}}(\omega)$ and the same overlap-save block performs CD compensation and matched filtering simultaneously. Alternatively the matched filter can be folded into the taps of the adaptive equaliser, provided the adaptive FIR is long enough to span both the matched-filter response and the channel memory it tracks; this saves the explicit filter but requires a longer adaptive FIR.

3.2.5 Filter Length

The CD equaliser must span the channel memory introduced by dispersion. An experimentally verified length for several pulse shapes is given by [10]

$$N_{\text{CD}} = \left\lceil \frac{6.67|\beta_2|LR_s}{T_s} \right\rceil, \quad (3.10)$$

where T_s is the receiver sampling period and $\beta_2 = -\lambda^2 D / (2\pi c)$ is the group-velocity dispersion parameter. With an oversampling factor of $\eta = 2$ and the worst-case CPON tolerance of 1600 ps/nm at 1550 nm ($|\beta_2| \approx 20.4 \text{ ps}^2/\text{km}$), this yields a 22-tap filter. The associated overlap-save FFT size is chosen as the power of two that minimises the per-symbol multiplication count, $N = 128$.

The adaptive equaliser must span the channel memory introduced by PMD. The mean differential group delay is $\langle \Delta\tau \rangle = D_{\text{PMD}}\sqrt{L}$, and the instantaneous DGD follows a Maxwellian distribution [15]

$$\mathbb{P}[\Delta\tau > x] \approx \frac{4}{\pi} \frac{x}{\langle \Delta\tau \rangle} \exp\left(-\frac{4x^2}{\pi\langle \Delta\tau \rangle^2}\right). \quad (3.11)$$

The required span at an outage probability of 10^{-12} and the CPON tolerance of 8 ps total PMD is well under one sampling period $T_s = 16.4 \text{ ps}$. The adaptive equaliser can therefore collapse to a single tap per cross-term, performing only polarisation demultiplexing once CD compensation and matched filtering have been done upstream.

3.2.6 Computational Complexity

A radix-2 FFT of size N decomposes into $\log_2 N$ stages of $N/2$ butterflies, each costing 4 real multiplications and 6 real additions. Each overlap-save block needs a forward FFT, an inverse FFT of equal cost, and N complex multiplications by the precomputed filter

spectrum, shared over $N - N_{\text{CD}} + 1$ valid outputs. The FFT can be simplified further by rounding each twiddle factor to the nearest signed power of two in its real and imaginary parts, replacing every twiddle multiplication with a pair of bit-shifts and eliminating the real multiplications inside the FFT and IFFT entirely. The per-symbol counts of the time-domain FIR, overlap-save, and overlap-save with power-of-two twiddles are summarised in table 3.1, where $\eta = f_s/R_s$ is the receiver oversampling factor.

Implementation	RM / symbol	RA / symbol
Time-domain FIR	$4\eta N_{\text{CD}}$	$\eta(4N_{\text{CD}} - 2)$
Overlap-save (FD)	$\eta \frac{4N \log_2 N + 4N}{N - N_{\text{CD}} + 1}$	$\eta \frac{6N \log_2 N + 2N}{N - N_{\text{CD}} + 1}$
Overlap-save with 2^k twiddles	$\eta \frac{4N}{N - N_{\text{CD}} + 1}$	$\eta \frac{6N \log_2 N + 2N}{N - N_{\text{CD}} + 1}$

Table 3.1: Per-symbol real multiplication (RM) and real addition (RA) counts for the three CD equaliser implementations. N is the overlap-save FFT size, N_{CD} the filter length from eq. (3.10), and η the receiver oversampling factor.

The per-symbol cost of the adaptive equaliser is dominated by the four FIR outputs and the weight update, which both scale with the filter length N_{AEQ} . The direct update of eq. (3.6) costs four complex multiplications per tap; the sign-sign reformulation of section 3.2.3 replaces these by sign manipulations and is multiplier-free. Per-symbol counts for both variants are given in table 3.2; the parallelisation of section 3.2.3 amortises the single update over P symbols and so does not appear in the per-symbol count.

Stage	RM / symbol	RA / symbol
FIR outputs ($4 \text{ FIRs} \times N_{\text{AEQ}} \text{ taps}$)	$8N_{\text{AEQ}}$	$4N_{\text{AEQ}} + 2$
Error terms	2	1
Weight update – direct	$8N_{\text{AEQ}} + 2$	$8N_{\text{AEQ}}$
Weight update – sign-sign	0	$8N_{\text{AEQ}}$

Table 3.2: Per-symbol real-operation counts for the parallel-lane butterfly CMA equaliser with N_{AEQ} taps per FIR. Sign-sign refers to the multiplier-free update described in section 3.2.3.

3.3 Clock Recovery

3.3.1 Gardner Timing Recovery

The Gardner TED computes an error term from samples $x[n]$ at $\eta = 2$ as [22]

$$e[k] = \Re\{x^*[2k-1](x[2k] - x[2k-2])\}. \quad (3.12)$$

The sample $x[2k-1]$ sits midway between two symbol-rate samples; when timing is locked, this midpoint vanishes and $e[k] = 0$, while a small timing error makes the midpoint take the sign of the slope between $x[2k-2]$ and $x[2k]$. The error feeds a proportional-integral

(PI) DPLL whose output drives a cubic Farrow interpolator [6] that re-samples the input. The loop uses the same P -lane parallelisation as the adaptive equaliser (section 3.2.3): each lane runs its own interpolator with a per-lane fractional interval, but all lanes within a block share the same PI output so the TED contributions of every lane can be summed before the loop filter is updated.

3.3.2 Modified Godard Timing Recovery

An RRC pulse with symbol period $T_{\text{sym}} = 1/R_s$ has bandwidth $(1 + \beta)/T_{\text{sym}}$, so for the Fourier transform of the received signal $X(f)$ the product $C(f) = X(f)X^*(f - 1/T_{\text{sym}})$ is non-zero only where the spectra overlap, and is real-valued in this region due to the symmetry of the RRC spectrum. A timing offset τ introduces a phase ramp in the frequency domain, so the overlap becomes

$$C(f) = X(f)X^*\left(f - \frac{1}{T_{\text{sym}}}\right) e^{-j2\pi\tau/T_{\text{sym}}}, \quad (3.13)$$

whose phase is independent of f and encodes the timing offset. A discrete equivalent for oversampling factor η is given by [12]

$$S = \sum_{k=k_{\text{lo}}}^{k_{\text{hi}}} X[k] X^*[k + (1 - 1/\eta)N], \quad k_{\text{lo}} = \frac{1-\beta}{2\eta}N, \quad k_{\text{hi}} = \frac{1+\beta}{2\eta}N, \quad (3.14)$$

where the summation is restricted to the bins covered by the RRC excess band, with k_{lo} and k_{hi} rounded to the nearest integer. The timing error normalised to the sampling period is then

$$\hat{\tau}_s = -\frac{\eta}{2\pi} \arg(S), \quad (3.15)$$

which is fed into a PI loop filter; the next block applies the accumulated τ_s as a frequency-domain phase ramp on the FFT output before the IFFT. Because S is formed from the same FFT used by the overlap-save CD/matched-filter stage, the Modified Godard TED adds no additional FFT cost.

3.3.3 Computational Complexity

The two detectors operate on blocks of different sizes – Gardner on a block of P output samples at $\eta = 2$ (i.e. $P/2$ symbols), Modified Godard on an overlap-save block of N FFT bins producing $N - N_{\text{CD}} + 1$ valid samples (i.e. $(N - N_{\text{CD}} + 1)/\eta$ symbols) – so the per-block counts are normalised to per-symbol counts in table 3.3, counting complex multiplications as 4 RM + 2 RA and complex additions as 2 RA. The Gardner cost is dominated by one cubic Farrow interpolator per output sample, evaluated as four fixed-coefficient 4-tap FIRs on complex data (one per polynomial coefficient $c_0, \dots, c_3$) followed by evaluation of the polynomial in μ , giving 38 RM and 30 RA per output sample. The Godard cost is dominated by the $\beta N/\eta$ complex products of eq. (3.14) and the per-block frequency-domain phase ramp (N recursive ramp coefficients and N spectral-bin multiplications, $\approx 8N$ RM); both are amortised over the $(N - N_{\text{CD}} + 1)/\eta$ valid symbols delivered by the overlap-save block. The FFT and IFFT used by Godard are reused from the overlap-save CD/matched-filter stage and so appear in table 3.1 rather than here.

Stage	RM / symbol	RA / symbol
Gardner TED	4	8
Gardner Farrow interpolator	76	60
Godard metric S (eq. (3.14))	$\frac{4\beta N}{N - N_{\text{CD}} + 1}$	$\frac{4\beta N}{N - N_{\text{CD}} + 1}$
Godard $\arg(S)$ (CORDIC)	$\frac{\eta}{N - N_{\text{CD}} + 1}$	0
Godard FD phase ramp	$\frac{8N\eta}{N - N_{\text{CD}} + 1}$	$\frac{4N\eta}{N - N_{\text{CD}} + 1}$

Table 3.3: Per-symbol real-operation counts for the two clock-recovery detectors. Gardner runs at $\eta = 2$ and shares the parallelisation of section 3.2.3; the per-lane Farrow and TED costs collapse to a fixed per-symbol count after dividing by $P/2$ symbols per block. Modified Godard runs once per overlap-save block of N samples, amortising over $N - N_{\text{CD}} + 1$ valid output samples (the overlap-save overhead). The shared FFT/IFFT used by Godard is accounted for in table 3.1.

3.3.4 Fixed-Point

The loop filter used in both clock recovery algorithms requires care when implemented in fixed-point arithmetic. The proportional and integral constants k_p, k_i can be very small so the loop filter output will be quantised to 0 if it shares the same dynamic range as the error estimate. This can be corrected by choosing k_p, k_i to be powers of 2 such that the quantised error terms can be bit-shifted then combined to give the loop-filter output $W[n] = k_p e[n] + k_i \sum_{l=0}^{n-1} e[n-l]$. This requires $W[n]$ to be a high precision accumulator, but the energy consumption from bits changing in the accumulator is determined by the precision of the error terms. To perform energy-efficient timing correction, $W[n]$ is quantised to low precision before being used either in the phase ramp for Godard or Farrow interpolator for Gardner where energy consumption must be minimised. The FFT scales the input by $1/N$ so Godard can end up estimating the timing error from small values. This can be remedied by applying a left bit-shift that approximates multiplying by $\sqrt{N}$, effectively making the forward FFT have unit gain, before quantisation at the precision of the timing error estimate. Then a right bit-shift is applied to undo this before being scaled by k_i and k_p in the loop filter.

3.4 Combined Implementations

A common architecture for short-range access-network receivers uses a single butterfly adaptive equaliser to perform all equalisation roles in one 2×2 MIMO structure [13]. For a longer network like the one under investigation, however, this filter must be at least $N_{\text{CD}} = 22$ taps long (eq. (3.10)), or up to $22 + 2 \times 10 - 1 = 41$ taps for a conservative equaliser that fully compensates CD and the RRC span.

At those lengths a time-domain FIR is no longer the cheapest implementation, and both the static and the adaptive equalisers become candidates for a frequency-domain implementation. Replacing the static CD/matched-filter FIR with an overlap-save FFT block drops the per-symbol multiplication count from $O(N)$ to $O(\log_2 N)$ (table 3.1), and the

resulting FFTs are then available for the Modified Godard TED. A frequency-domain adaptive equaliser [8] achieves the same per-symbol FFT scaling, but the MIMO structure is wasteful when PMD is negligible. The Modified Godard TED also cannot be placed after it as the output would be at the symbol rate.

This motivates two combined implementations. An overlap-save static equaliser performs CD compensation and matched filtering, followed by either Modified Godard (in the frequency domain) or Gardner (after the IFFT) and finally a single-tap adaptive equaliser for polarisation demultiplexing.

3.5 Methodology

Each combined implementation is evaluated on a DP-QPSK channel sized to the worst-case CPON tolerances of section 1.3.3. The channel comprises 80 km of fibre at $D = 20$ ps/(nm km) (1600 ps/nm of CD), $\beta = 0.25$ RRC pulse shaping at $\eta = 2$, PMD with $D_{\text{PMD}} = 0.1$ ps/ $\sqrt{\text{km}}$ drawn from a fixed seed (held constant across every trial, SNR, and CFO), a 40 ppm sample-frequency offset, and AWGN at a swept SNR from 0 to 30 dB. The adaptive equaliser uses a step size of $\mu = 2^{-10}$. Each trial uses $N_s = 18750$ symbols per polarisation, enough for 1.5 samples of timing drift at 40 ppm to exercise the clock recovery, and BER is estimated by Monte Carlo over 10 trials. The RSNR is found per trial by interpolation of $\log_{10}(\text{BER})$ between adjacent SNR samples and reported as a mean with standard deviation. The PI gains (k_i, k_p) of the Gardner and Modified Godard loops are pre-tuned by a logarithmic grid search, then held for the full sweep. Two design choices are varied across the comparison: the static-equaliser length (22 taps to compensate CD only, or 41 taps to also span the matched-filter response) and whether the FFT/IFFT twiddle factors are rounded to a power of two.

An initial algorithm comparison is performed in floating-point arithmetic to determine the effects of CFO and to give benchmark values for the fixed-point implementation.

Coarse CFO correction. Because carrier recovery sits downstream of equalisation and clock recovery, a residual CFO is present at this stage and degrades both timing recovery algorithms (fig. 3.1). An energy-efficient coarse correction can be built from the FFT already produced by the overlap-save CD/matched-filter stage. The CFO estimate is found from the centroid of the FFT magnitudes, averaged over multiple blocks to improve accuracy:

$$\hat{k} = \frac{\sum_k k |X[k]|^2}{\sum_k |X[k]|^2}, \quad k \in \{0, \dots, \frac{N}{2} - 1, -\frac{N}{2}, \dots, -1\}, \quad (3.16)$$

with bin spacing $\eta R_s/N$. The estimate is applied as a continuous-valued time-domain phasor $\exp(-j2\pi \hat{k}n/N)$ to later samples before they reach the FFT. Estimating the CFO requires only an average over a small number of blocks, and applying the phasor requires only one CORDIC operation, so its energy cost is negligible compared to equalisation and clock recovery. The estimator is placed before the matched filter, whose pass-band is centred at DC and would otherwise bias the centroid towards zero.

Fixed-point. The fixed-point precision is defined as the number of bits used to represent fractional values. The input signal is limited to the range $[-1, 1]$ so the precision captures the full signal range. Pilots are rescaled down to the same range for this analysis as

they are not used and would otherwise require additional integer bits, as discussed in chapter 2. Equaliser values are also normalised to $[-1, 1]$, and the FFT/IFFT applies appropriate rescaling to avoid overflow. Three independently swept precisions are defined over $[-1, 1]$: one shared by the FFT/IFFT and static equaliser, one for the gradient terms of the adaptive equaliser, and one for the clock-recovery error estimate and correction. Each precision is first swept individually while the others are held at high precision, to determine the lowest viable precision of each stage. The resulting low precisions are then combined to give suitable operating points for use in the full DSP pipeline.

3.6 Results & Discussion

3.6.1 Combined Performance

Table 3.4 compares the performance of the proposed implementations at floating-point precision in the absence of CFO. The conservative static equaliser length of 41 offers little improvement over 22 taps, so 22 taps can be used to reduce complexity. Rounding the FFT/IFFT twiddle factors to a power of 2 incurs an SNR penalty but is still worth considering, since the receiver’s SNR floor may be determined by other DSP blocks.

In reality a CFO can be present in the signal, as the carrier recovery block (chapter 4) is downstream of equalisation and clock recovery. The effect of CFO on performance is shown in fig. 3.1. CFO is clearly problematic for both clock recovery algorithms, with neither able to achieve the 3 GHz tolerance, but the Godard TED struggles the most. The issue can be seen from eq. (3.14); the summation used to perform the estimate is over specific frequency bins but a CFO shifts the spectrum and the sum no longer captures the intended overlap. The effect of a CFO Δf on Gardner is more subtle but can be seen by expressing the signal with CFO as $y[k] = x[k] \exp(j2\pi\Delta f T_s \eta k)$ then substituting into eq. (3.12) to give

$$e[k] = \Re \{x^*[2k-1]x[2k] \exp(j2\pi\Delta f T_s \eta) - x^*[2k-1]x[2k-2] \exp(-j2\pi\Delta f T_s \eta)\}. \quad (3.17)$$

Taking expectations and noting that the correlation of QPSK symbols is real, we obtain

$$\mathbb{E}[e[k]] = \cos(2\pi\Delta f T_s \eta) \{\mathbb{E}[x^*[2k-1]x[2k]] - \mathbb{E}[x^*[2k-1]x[2k-2]]\}, \quad (3.18)$$

so a CFO flattens $\mathbb{E}[e[k]]$ (known as the S-curve), making it harder for the loop to track errors.

Block	N_{CD}	N_{AEQ}	2^k twiddles	RSNR (dB)
Gardner	22	1	no	3.46 ± 0.08
Gardner	22	1	yes	9.89 ± 0.76
Gardner	41	1	no	3.40 ± 0.06
Gardner	41	1	yes	9.83 ± 0.68
Godard	22	1	no	3.42 ± 0.09
Godard	22	1	yes	7.00 ± 0.38
Godard	41	1	no	3.66 ± 0.09
Godard	41	1	yes	6.50 ± 0.11

Table 3.4: RSNR (mean $\pm$ standard deviation over trials) for each combined equalisation/clock recovery configuration in the absence of CFO. N_{CD} and N_{AEQ} are the number of taps in the static CD/matched-filter and the adaptive equaliser respectively.

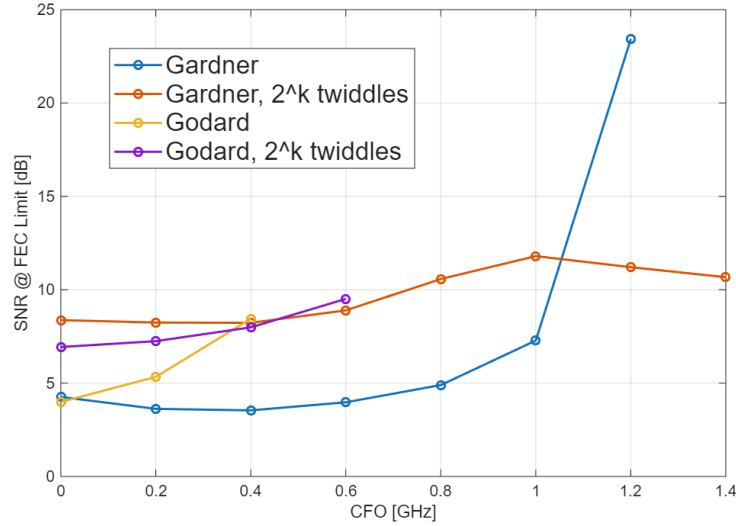


Figure 3.1: RSNR as CFO varies for each implementation. Points are omitted when the FEC limit is no longer reached at any SNR.

Correcting CFO Figure 3.2 shows the RSNR for each implementation once the coarse CFO correction of section 3.5 is applied. The coarse estimate removes enough CFO for all implementations to achieve the FEC limit at the 3 GHz tolerance. Godard and Gardner with the standard FFT/IFFT exhibit the same performance, but Godard performs better once the 2^k twiddle factors are included. Rounding the twiddle factors impacts both the FFT and IFFT; as Godard operates on data after the FFT, it is not affected by the additional loss of accuracy in the IFFT like Gardner.

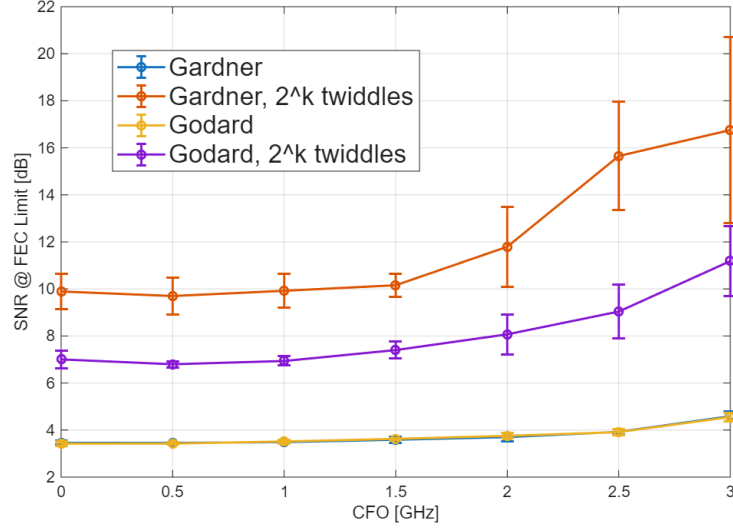


Figure 3.2: RSNR as CFO varies with coarse CFO correction.

3.6.2 Fixed-Point Precision

So far implementations have been compared in floating-point arithmetic, avoiding the quantisation error present in a real coherent receiver. Each stage will have different precision requirements, so a search over precision combinations is required.

Adaptive equaliser. Figure 3.3 shows how the precision of the adaptive equaliser gradient estimate impacts performance for each implementation while the static equaliser and clock recovery operate at high precision. Unexpectedly, performance improves across all implementations as the precision is reduced down to 4 bits before failing at 2 bits. The reduction in RSNR at lower precision when 2^k twiddle factors are used is enough to be significant over the 5 runs; a possible explanation is that at high static equaliser and clock recovery precision, the only additional error introduced is by the twiddle factors

Static equaliser. Figure 3.4 shows the same precision sweep for the precision of the FFT/IFFT and the static filter when clock recovery and the adaptive equaliser are held at high precision. The precision requirement is slightly higher here but Godard and Gardner perform well down to 8 bits of precision. This can be explained by the FFT implementation: to avoid overflow each stage of the radix-2 FFT applies a $1/2$ scaling with a bit-shift. For an FFT of size 128 this is 7 bit-shifts which at a precision lower than 8 bits means that the least significant bit is guaranteed to be lost at each stage, resulting in a very large quantisation error.

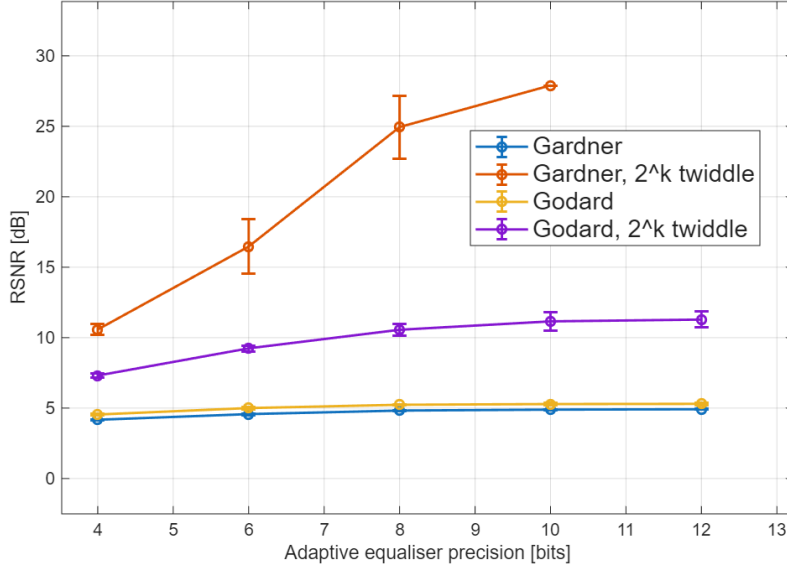


Figure 3.3: RSNR as the fixed-point precision of adaptive equalisation varies, with static equalisation and clock recovery at high precision.

Clock recovery. Figure 3.5 shows the sweep for the precision of the timing error estimate and correction (Farrow interpolation or phase ramp). The loop filter is kept at high precision, as are the static and adaptive equalisers. Without the 2^k twiddle factors, Godard and Gardner perform the same down to 6 bits of precision before a sharp rise in RSNR at 4 bits, with Gardner failing to reach the FEC limit. Godard clearly outperforms Gardner when 2^k twiddle factors are used, as in the floating-point case. The Godard estimator is vulnerable to fixed point errors when computing the estimation metric from eq. (3.14) as it requires a sum over $\beta N/\eta = 16$ terms where quantisation error can compound more than in the simple Gardner estimate of eq. (3.12). However, Gardner is more susceptible to fixed-point precision at the correction stage where the Farrow interpolator performs many more cascaded operations than CORDIC.

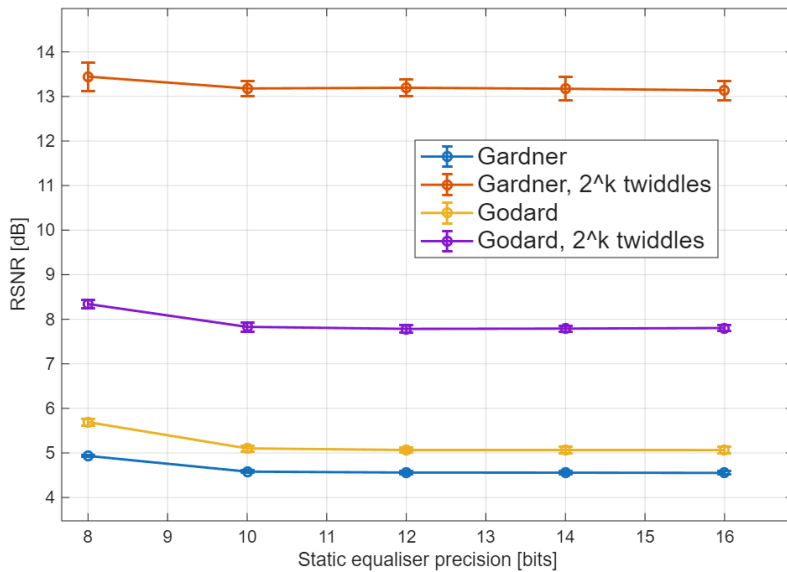


Figure 3.4: RSNR as the fixed-point precision of static equalisation varies, with adaptive equalisation and clock recovery at high precision.

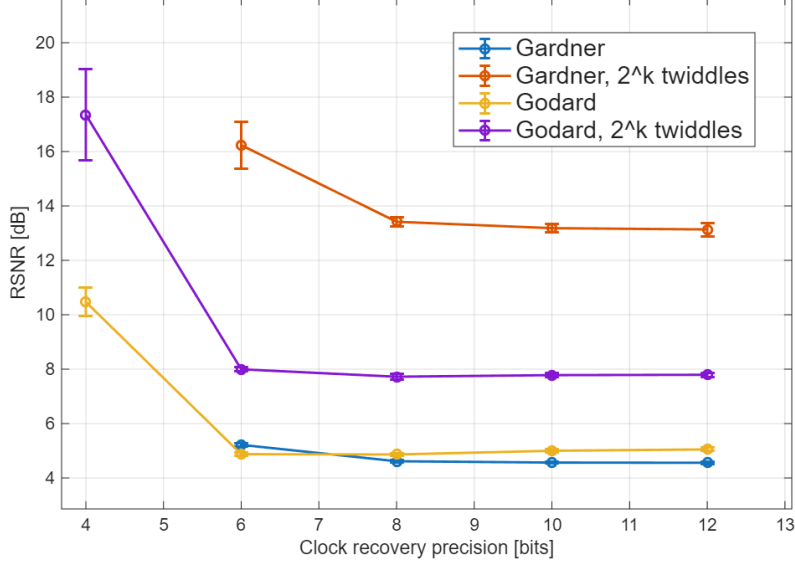


Figure 3.5: RSNR as the fixed-point precision of clock recovery varies, with static and adaptive equalisation at high precision.

Low precision operating points. Combining each stage at low precision can introduce cascaded quantisation error which incurs an SNR penalty. Table 3.5 shows the RSNR and energy breakdown of each algorithm at the lowest feasible precision combination taken from each individual sweep as well as a slightly higher precision combination for comparison. Godard achieves a lower RSNR than Gardner across all implementations at a lower energy. The choice of Godard implementation then depends on the balance of receiver energy and how the RSNR affects the receiver energy. A suitable low-energy implementation is to use 2^k twiddle factors with a precision combination of (8,6,6) as the slight increase in precision significantly improves the RSNR over the (6,4,4) implementation. If a low RSNR is needed then foregoing the 2^k twiddle factors at a precision of (6,4,4) is the most attractive as the SNR penalty compared to the higher precision implementation is negligible but saves energy. These two implementations will be taken forward to the full pipeline analysis in chapter 5.

Eq.	2^k	Precision (bits)			RSNR (dB)	Energy (pJ/bit)			
		Stat.	Adapt.	Clk		E_{stat}	E_{aeq}	E_{clk}	E_{tot}
Gardner	No	6	4	4	7.91 ± 0.68	15.00	1.64	12.70	29.33
		8	6	6	4.80 ± 0.27	17.76	1.97	15.30	35.03
Gardner	Yes	6	4	4	—	2.68	1.64	12.70	17.01
		8	6	6	20.60 ± 2.04	3.09	1.97	15.30	20.37
Godard	No	6	4	4	4.80 ± 0.18	15.00	1.64	3.18	19.81
		8	6	6	4.12 ± 0.19	17.76	1.97	3.84	23.56
Godard	Yes	6	4	4	10.02 ± 0.53	2.68	1.64	3.18	7.49
		8	6	6	7.77 ± 0.47	3.09	1.97	3.84	8.90

Table 3.5: Fixed-point performance and energy of the combined clock recovery and equalisation implementations at low- (6/4/4) and high-precision (8/6/6) operating points, with and without 2^k twiddle factors. The three precisions are those of the static equaliser, adaptive equaliser and clock recovery stages. The Gardner low-precision case with 2^k twiddle factors fails to reach the FEC limit at any SNR. Proposed candidates for an energy-efficient receiver are highlighted in bold.

Oversampling rate. These results assume an oversampling rate of 2. Modern coherent receivers can operate below 2 samples per symbol to reduce the load on analogue-to-digital converters (ADCs) and simplify the DSP [20]. The Godard scheme allows the full equalisation and clock recovery block to operate at a fractional oversampling rate which reduces the number of operations per symbol. However, Gardner requires 2 samples per symbol so the output of the IFFT would need to be upsampled and the savings from a lower oversampling rate would be smaller.

Chapter 4

Carrier Recovery

4.1 Introduction

A coherent receiver recovers the transmitted symbols by mixing the incoming optical signal with a local oscillator (LO) laser, allowing the amplitude and phase of each symbol to be extracted directly. However, the transmitter and LO lasers generally have different centre frequencies and independent phase trajectories, producing a carrier frequency offset (CFO) and accumulated phase noise on the received symbols. Carrier recovery is the DSP stage responsible for estimating and removing these impairments before symbol decisions are made.

Carrier recovery is divided into two sequential stages. Frequency recovery performs a finer estimate of the CFO than that discussed in chapter 3 and corrects it so that only slow residual phase variations remain. The frequency recovery discussed in this chapter is intended to be agnostic to whether the coarse CFO correction described there has already been applied, so as to generalise across receiver designs; the worst-case CFO of 3 GHz is therefore assumed. Phase recovery then tracks and removes these residual phase errors on a block-by-block basis.

Two common carrier recovery algorithms, Rife & Boorstyn and Viterbi & Viterbi, are compared against low-complexity alternatives to quantify the trade-offs between energy consumption and performance in an energy-efficient carrier recovery block.

4.2 Frequency Recovery

4.2.1 Carrier Frequency Offset

A carrier frequency offset (CFO) occurs because the transmitter and local-oscillator lasers are free-running and never exactly aligned in centre frequency. The resulting beat appears as a linear phase ramp on the received samples. It is simulated by multiplying the signal by $\exp(j 2\pi \Delta f T_{\text{sym}} k)$, where Δf is the frequency offset, $T_{\text{sym}} = 1/R_s$ the symbol period (carrier recovery operates at one sample per symbol), and k the symbol index.

4.2.2 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4} + m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i + \theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i + 4\theta[i])} + n'[i] \quad (4.1)$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise, which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i + \theta[i])} + n[i], \quad (4.2)$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$, which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4 \arg(x[i])\}|_{-\pi}^{\pi} \quad (4.3)$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \quad (4.4)$$

for data-aided operation, where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

4.2.3 Rife & Boorstyn

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \dots, z[i+L]$ where L is the observation length. The Rife & Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give [11]

$$\delta = -\Re \left[\frac{X_{k+1} - X_{k-1}}{2X_k - X_{k-1} - X_{k+1}} \right]. \quad (4.5)$$

The updated index $k' = k + \delta$ is then scaled by the sampling rate to find the true CFO. The resolution in k is $\epsilon(k) = 1/2N_{FFT}$. The CPON specification provides only $L = 11$ training symbols, so data-aided operation without zero-padding gives $\epsilon(\hat{f}_d) \approx 0.05 R_s$ — too coarse for the downstream phase recovery to correct. Zero-padding the training sequence to a larger N_{FFT} improves the resolution without requiring additional training data. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 4.1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarisations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) provides a lower bound on the achievable MSE and is given by [4]

$$\text{MCRB} = \frac{3}{2\pi^2 L^3 \times \text{SNR}}. \quad (4.6)$$

Rife & Boorstyn achieves this bound above a threshold SNR of ≈ 4 dB, where the threshold SNR is defined as the SNR below which the estimator begins to deviate from the

MCRB. Below this, noise contributes significantly to the FFT values and can lead to an incorrect index being chosen.

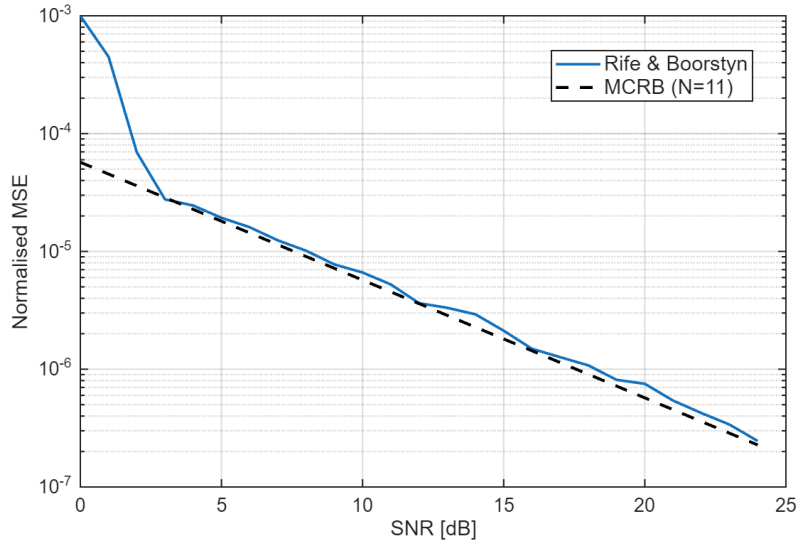


Figure 4.1: Normalised MSE of the Rife & Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the Rife & Boorstyn algorithm improves with longer observation lengths, as shown in fig. 4.2 for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

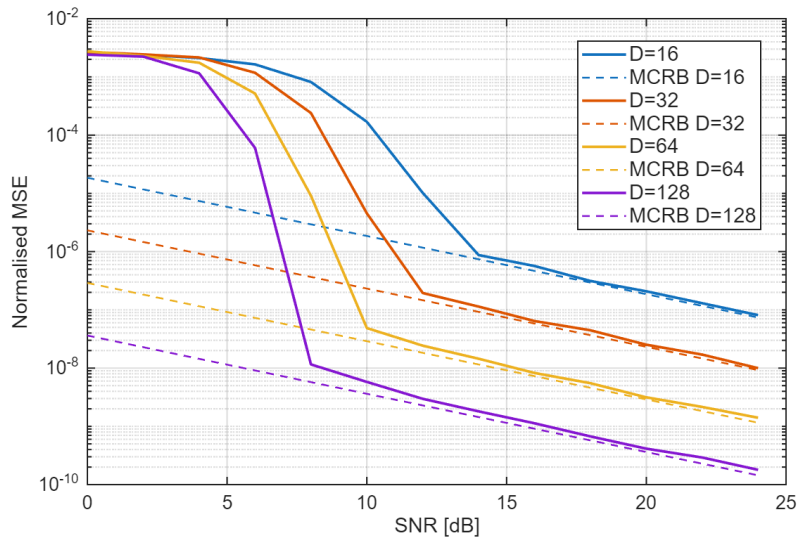


Figure 4.2: Normalised MSE of the Rife & Boorstyn algorithm for non-data-aided operation and different observation lengths.

4.2.4 Differential Phase and Kay Estimator

The Rife & Boorstyn algorithm exhibits good performance even at low SNRs, but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2} \log_2(N_{FFT})$ complex multiplications

and $N_{FFT} \log_2(N_{FFT})$ complex additions. The fine search from eq. (4.5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_{\text{sym}} + \theta[i] - \theta[i-1] + \eta[i] \quad (4.7)$$

over several symbols and averaging to remove the effect of θ . Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$, so it exhibits poor accuracy. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi} \sum_{k=1}^L \frac{6k(L-k)}{L(L^2-1)} \arg(z[k]z^*[k-1]). \quad (4.8)$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (4.7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB, but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. When the total phase drift $\Delta\phi = 2\pi f_d L T_{\text{sym}}$ across the observation window exceeds 2π , a wrap shifts the estimated CFO by $\pm 1/(L T_{\text{sym}})$. A first-pass coarse CFO correction must therefore be applied before the Kay estimator to keep the residual CFO within the unambiguous range. The performance of the differential phase and Kay estimator for data-aided operation is shown in fig. 4.3. The MCRB is achieved above a threshold SNR of ≈ 7 dB. This is higher than for Rife & Boorstyn, but at a reduced computational cost. Unlike the Rife & Boorstyn algorithm, the threshold SNR of the differential phase and Kay estimator does not depend on observation length, as shown in fig. 4.4.

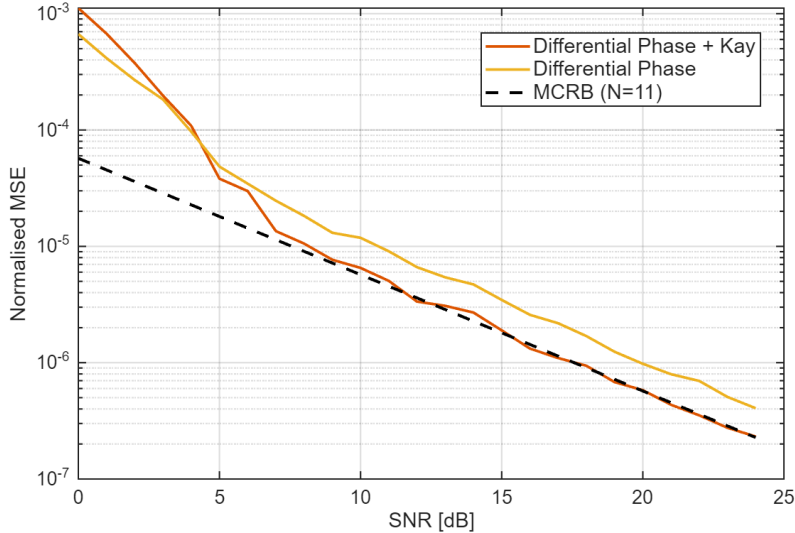


Figure 4.3: Normalised MSE of frequency estimation with the differential phase and Kay estimator for $L = 11$ training symbols.

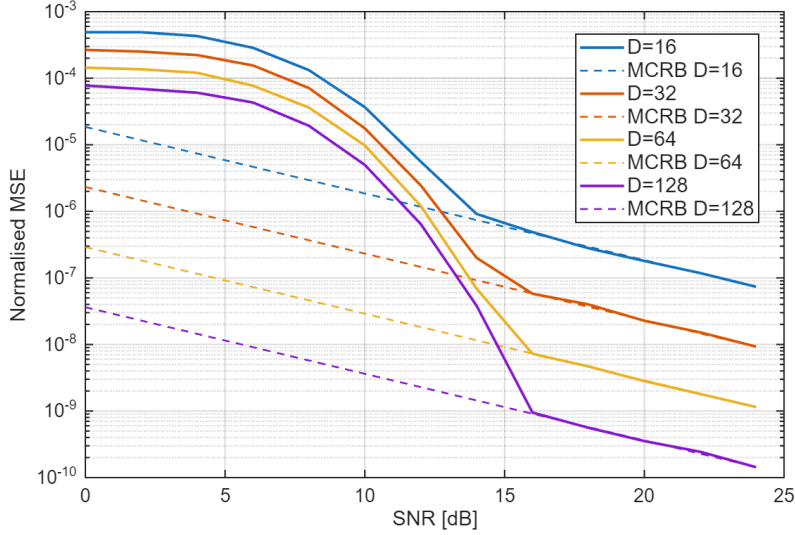


Figure 4.4: Normalised MSE of the differential phase and Kay estimator with blind operation for different observation lengths.

4.2.5 Computational Complexity

Here, the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. The real division required by the Rife & Boorstyn parabolic interpolation is assumed to cost the same as a multiplication since it is only performed once per estimate; a look-up-table reciprocal would meet this approximation at a small accuracy cost. All analysis applies to a single polarisation. If averaging is done over both polarisations, then all operations are doubled.

Rife & Boorstyn The Rife & Boorstyn algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re \left[\frac{a + bj}{c + dj} \right] = \frac{ac + bd}{c^2 + d^2}. \quad (4.9)$$

The total complexity is summarised in table 4.1; entries marked with a dagger (†) apply only to blind operation.

Operation	RM	RA
Forming $z[i], \dots, z[i + L]$	$4L, 12L^\dagger$	$3L, 9L^\dagger$
FFT	$2L \log_2(\frac{N_{FFT}}{L}) + 2N_{FFT} \log_2(L)$	$3L \log_2(\frac{N_{FFT}}{L}) + 3N_{FFT} \log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 4.1: Computational complexity of estimating CFO with the Rife & Boorstyn algorithm.

Differential phase and Kay The differential phase and Kay estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \dots, \arg(z[i + L])$	$2L$	L
Differential Phase	1	$L - 1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \dots, \arg(z[i + L])$	$2L$	L
Kay	L	$L - 1$

Table 4.2: Computational complexity of estimating CFO with the differential phase and Kay estimator.

4.2.6 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s . For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} for an estimation range $0 \leq |f_d/R_s| \leq 1$. In reality, a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s|$ is 2^{-d} , then d bits of precision that would otherwise be redundant can be reclaimed by scaling each estimate by 2^d during calculation and inserting d bits into the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

4.2.7 Maximum CFO

The theoretical limit on the maximum CFO that can be corrected is given by Nyquist as $0.5 R_s$. From eq. (4.1), we see that blind operation reduces this limit by a factor of 4. Both the Rife & Boorstyn and differential phase and Kay estimators accurately estimate the CFO up to the Nyquist limit, as seen in fig. 4.5. Blind operation aliases as expected at $f/R_s = 0.125$, making it viable for CFOs up to 3.8 GHz at a symbol rate of 30.5 GBd.

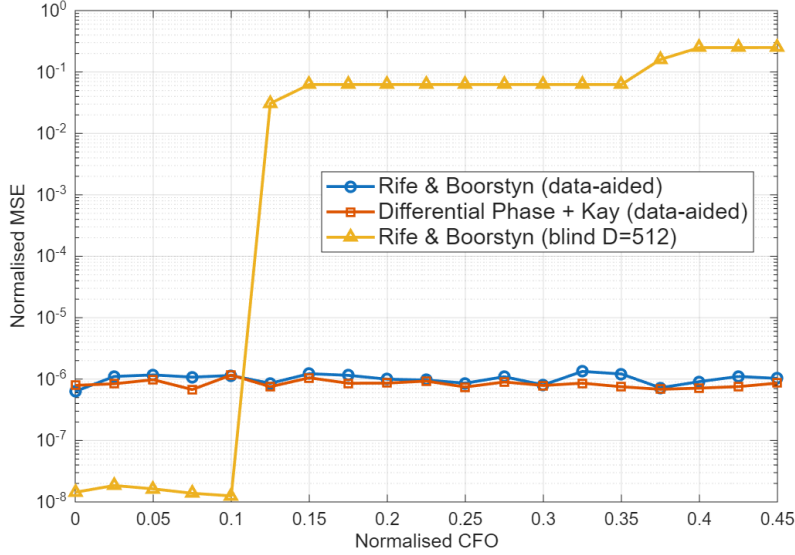


Figure 4.5: Normalised MSE in CFO estimate (MSE/R_s^2) vs. normalised CFO (f/R_s).

4.3 Phase Recovery

4.3.1 Phase Noise

Phase noise originates from the finite linewidth of the transmitter and LO lasers, whose phases follow a random walk. It is modelled as a Wiener process: independent Gaussian per-symbol phase increments with variance $\sigma^2 = 2\pi \Delta\nu/R_s$, where $\Delta\nu$ is the combined laser linewidth, are accumulated and applied as a multiplicative phase $\exp(j\phi[k])$.

4.3.2 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small-angle approximations. The CPON specification sets the symbol rate at 30.5 GBd, so at an oversampling rate of 2 the receiver runs at 61 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of, e.g., 32 symbols, reducing the required clock speed to roughly 1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data are used instead.

4.3.3 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of eq. (4.1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg \left(\sum_{k=1}^N w_{ML}^*[k] z[k] \right) - \frac{\pi}{4}. \quad (4.10)$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied to all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON specification provides one pilot symbol in every 32-symbol block that can be used to correct this [3]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{\text{PU}}$ (the output of the phase unwrapper) and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{\text{PU}} = \hat{\theta}_{\text{PU}} - n\frac{\pi}{2}$.

4.3.4 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using eq. (4.2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm\pi/4$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

4.3.5 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 4.3.

Operation	RM	RA
Forming $z[i], \dots, z[i + N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N + 1$	$3N + 2(N - 1) + 1$

Table 4.3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in table 4.4.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 4.4: Computational complexity of Pilots-Only correction.

4.4 Methodology

Phase recovery comparison. The two phase recovery algorithms are compared in isolation by simulating an AWGN channel with additive Wiener phase noise, without

chromatic dispersion, PMD, or fibre nonlinearity, so that any performance difference is attributable solely to the phase estimation stage. BER is estimated by Monte Carlo simulation over 5 independent realisations for each combination of laser linewidth and SNR, with the phase ambiguity inherent to QPSK resolved by exhaustive search over the four $\pi/2$ rotations before symbol decisions are made.

Frequency recovery comparison. Three frequency recovery configurations are compared by the receiver effort needed to estimate the CFO accurately enough for the downstream phase recovery: data-aided differential phase and Kay, blind Rife & Boorstyn with $D = N_{FFT} = 2048$ (the largest power-of-two FFT that fits within a 3712-symbol subframe), and data-aided Rife & Boorstyn with $N_{FFT} = 2048$. The channel applies a CFO drawn at random with 1 MHz Wiener phase noise and AWGN at 20 dB. Each algorithm produces one CFO estimate per subframe; the per-subframe estimates are averaged and the RMS error over 30 random CFO/noise realisations is recorded as a function of the number of averaged subframes N . For each algorithm the smallest N that brings the RMS error below the residual CFO the phase recovery can absorb (≈ 10 MHz, consistent with its linewidth tolerance in fig. 4.6) is found. A high fixed-point precision (32-bit word length, 16 fractional bits) is used so that quantisation is not the limiting factor, and all angle estimates and rotations are performed with CORDIC.

Operation counts. The receiver effort of each algorithm is reported as the per-subframe real-multiplication (RM) and real-addition (RA) counts of tables 4.1 and 4.2, evaluated for the observation lengths above and multiplied by the number of subframes N required to reach the 10 MHz target.

4.5 Results & Discussion

4.5.1 Phase Recovery Performance

To replace Viterbi & Viterbi with the simplified pilots-only phase correction, it must exhibit comparable performance over the range of linewidths that could be expected during operation of the network. The BER for each phase recovery algorithm as SNR varies is shown in fig. 4.6. Both algorithms have almost identical performance (plots overlapping) and are reliable up to 10 MHz, well beyond the maximum of 1 MHz given in the CPON spec.

Since the two phase recovery algorithms perform almost identically over the full linewidth and SNR range, the simpler pilots-only correction is adopted for phase recovery.

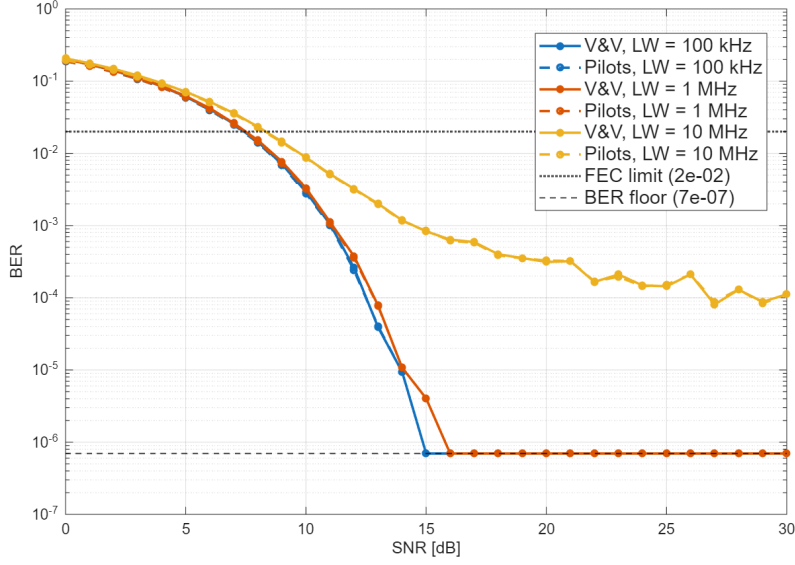


Figure 4.6: Performance comparison of V&V and Pilots-Only phase correction.

4.5.2 Frequency Recovery: Subframe Averaging

The data-aided algorithms form their CFO estimate from only the 11 CPON training symbols, so a single-subframe estimate is noisy. The estimate can be refined by averaging over several subframes, provided the CFO stays constant over that interval, but this multiplies the receiver effort. Table 4.5 reports, for each algorithm, the number of subframes N that must be averaged to reduce the CFO enough to allow phase recovery to correct the remaining error. Ideally this should be ≈ 1 MHz, the same as the phase noise, but from fig. 4.6 it seems that 10 MHz does not incur a significant RSNR penalty so should also be considered.

Blind Rife & Boorstyn reaches the 10 MHz target from a single subframe (residual 0.22 MHz): its 2048-symbol observation gives a very low-variance estimate, and the data-dependent component of its error averages out across the random data symbols. Data-aided differential phase and Kay needs 12 subframes (residual 0.79 MHz). Data-aided Rife & Boorstyn never reaches the target and its error floors at 17 MHz which suggests reusing the same training symbols biases the estimator.

Tightening the target to 1 MHz results in an inefficient operation count. Blind Rife & Boorstyn needs four subframes (294 932 RM and 360 464 RA), while differential phase and Kay, though still cheaper by operation count, would have to average over 1532 subframes, far longer than the CFO can likely be held constant, and is therefore infeasible. The differential phase and Kay estimator is the clear choice here with a very low operation count.

FR algorithm	Target	N	Per subframe		Total	
			RM	RA	RM	RA
Differential phase and Kay	10 MHz	12	78	42	936	504
(data-aided, $L = 11$)	1 MHz	1532			119 496	64 344
Rife & Boorstyn	10 MHz	1	73 733	90 116	73 733	90 116
(blind, $D = N_{FFT} = 2048$)	1 MHz	4			294 932	360 464
Rife & Boorstyn	10 MHz	—	18 481	25 637	—	—
(data-aided, $L = 11$, $N_{FFT} = 2048$)	1 MHz	—			—	—

Table 4.5: Subframes to average (N) and resulting operation counts (real multiplications RM, real additions RA) to reach a 10 MHz and a 1 MHz RMS CFO estimate. The per-subframe counts are target-independent. Data-aided Rife & Boorstyn reaches neither target (its error floors at 17 MHz).

4.5.3 Fixed Point

The most energy-efficient implementation of the differential phase and Kay estimator combined with pilot-based phase correction must select fixed-point values that balance energy consumption with RSNR.

Chapter 5

Full DSP Pipeline

5.1 Introduction

5.2 Analogue-Digital Converters (ADCs)

The optical front end mixes the received signal with the LO in a pair of 90° hybrids and balanced photodetectors, producing four baseband photocurrents: the in-phase and quadrature components of each polarisation. A bank of ADCs digitises these waveforms before any digital processing. At the CPON symbol rate of 30.5 GBd and the oversampling factor $\eta = 2$ used throughout this report, each converter runs at $\eta R_s \approx 61$ GSa/s. High-speed converters at these rates are among the most power-hungry components of a coherent receiver, and their resolution sets the precision floor of every downstream DSP block, so the number of bits is a primary energy/performance trade-off.

Quantisation-error SNR. A uniform quantiser of B bits spanning a full-scale range V_{FS} has a step size $\Delta = V_{\text{FS}}/2^B$. Provided the input is busy across many codes, the quantisation error e is well modelled as uniformly distributed over $[-\Delta/2, \Delta/2]$ and white, with variance

$$\sigma_e^2 = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} e^2 \text{d}e = \frac{\Delta^2}{12}. \quad (5.1)$$

For a full-scale sinusoid of amplitude $V_{\text{FS}}/2$ the signal power is $V_{\text{FS}}^2/8$, so the signal-to-quantisation-noise ratio is

$$\text{SQNR} = \frac{V_{\text{FS}}^2/8}{\Delta^2/12} = \frac{V_{\text{FS}}^2/8}{V_{\text{FS}}^2/(12 \cdot 2^{2B})} = \frac{3}{2} 2^{2B}, \quad (5.2)$$

or, in decibels,

$$\text{SQNR}_{\text{dB}} = 6.02 B + 1.76. \quad (5.3)$$

Each additional bit therefore buys roughly 6 dB of quantisation SNR.

Effective number of bits. A real converter falls short of eq. (5.3) because of thermal noise, sampling-clock jitter, comparator metastability, and non-linear distortion. These

are collapsed into a single figure of merit, the effective number of bits (ENOB), by measuring the signal-to-noise-and-distortion ratio (SINAD) for a full-scale sinusoidal input and inverting eq. (5.3):

$$\text{ENOB} = \frac{\text{SINAD}_{\text{dB}} - 1.76}{6.02}. \quad (5.4)$$

The ENOB is the resolution of an ideal quantiser that would introduce the same total noise power, and is always smaller than the nominal bit count B ; at the ~ 60 GSa/s rates required here, state-of-the-art converters achieve only around 5–6 effective bits. Two further considerations apply to the coherent signal of interest. First, the derivation above assumes a full-scale sinusoid: the pulse-shaped DP-QPSK waveform has a non-trivial peak-to-average power ratio, so the input must be backed off from full scale to avoid clipping its peaks, trading a few decibels of loading for reduced clipping distortion — the same balance captured by the 99.9th-percentile normalisation of chapter 2. Second, it is the ENOB rather than the nominal resolution that should be matched to the fixed-point word lengths of the DSP, since quantising the data more finely than the converter’s own noise floor expends energy for no gain in accuracy.

5.3 Orthogonalisation and Deskew

The quantisation model above assumes an ideal front end, but in practice the optical hybrid and the electrical paths feeding the ADCs are imperfect, leaving the in-phase and quadrature components mismatched. Three impairments are usually distinguished. A *gain imbalance* scales I and Q unequally; a *quadrature error* arises when the hybrid’s phase shift departs from 90° , so that the I and Q axes are no longer orthogonal; and an *I/Q timing skew* results from unequal propagation delays in the two paths, so that the I and Q samples correspond to slightly different instants in time.

Gain and quadrature errors are removed by *orthogonalisation*, classically the Gram–Schmidt orthogonalisation procedure (GSOP) [16, 17], which rescales the two components to equal power and subtracts the correlated part of one from the other to restore orthogonality. The timing skew is corrected by *deskew*, a fractional-delay interpolation — for example a Farrow structure like that used for clock recovery in chapter 3 — that realigns the I and Q sampling instants. Both are static or slowly varying impairments and are applied immediately after the ADC, ahead of equalisation.

These front-end corrections are not modelled in this report: the simulated channel delivers perfectly orthogonal, deskewed samples to the DSP so that the equalisation, clock-recovery, and carrier-recovery blocks can be characterised in isolation. They are noted here for completeness, since a hardware implementation would have to budget both for their (modest) energy cost and for the residual imbalance they leave behind, which adds to the SNR floor seen by the downstream blocks.

5.4 Methodology

5.5 Results

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